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A New Electroaffinity Scale; Together with Data on Valence States and on Valence Ionization Potentials and Electron Affinities

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A new approximate "absolute" scale of electronegativity, or electroaffinity, is set up. The absolute electroaffinity on this scale is equal to the average of ionization potential and electron affinity. These quantities must, however, in general, be calculated not in the ordinary way, but for suitable "valence states" of the positive and negative ion. Also, the electroaffinity of an atom has different values for different values of its valence; in general the electroaffinity as here calculated (in agreement with chemical facts) is larger for higher valences. Electroaffinity values have been calculated here for H, Li, B, C, N, O, F, Cl, Br, I. They show good agreement in known cases with Pauling's electronegativity scale based on thermal data, and with the dipole moment scale. The present electronegativity

scale (like the others) is rather largely empirical, especially as to its quantitative validity; and it remains to be seen whether or not the latter will be more than very rough when tested for a wider range of cases. Nevertheless the new scale has a degree of theoretical background and foundation which throws some new light on the physical meaning of the concept of electronegativity (or electroaffinity). The basis of the present scale, it should be mentioned, is simpler and more certain for univalent than for polyvalent atoms.—The nature of valence states of atoms is briefly discussed. It is hoped that the tabulations of atomic valence state energies and valence state ionization potentials and electron affinities given at the end of this paper may be useful in problems of molecular structure.

A. UNIVALENT ATOMS

ONE of the most familiar and useful of chemical concepts is that of relative electronegativity. The physical basis of this has remained obscure, although it has been evident that the electronegativity of an atom must be related somehow to its electron affinity or its ionization potential, or both. Recently two empirical scales of relative electronegativity have been set up. In the present paper, a possible third, absolute, scale is discussed.

In general, the wave function of a molecule AB, where A and B are both univalent, can be approximated by forming a linear combination of a wave function, which may be symbolized by $\psi(A-B)$, corresponding to a state with a Pauling-Slater electron-pair bond, with two wave functions corresponding to ionic states A^+B^- and A^-B^+ . That is, approximately,^{1, 2}

$$\psi_{AB} = \gamma\psi(A-B) + \alpha\psi(A^+B^-) + \beta\psi(A^-B^+). \quad (1)$$

If $\alpha = \beta$, we have a bond which, although it contains polar terms, is almost as purely nonpolar as any bond can be, and is a typical covalent bond.² If A and B are identical, $\alpha = \beta$ is obviously necessary; here γ is much larger than α or β .¹ For A not identical with B, Pauling states² that "if

the two atoms have the same degree of electronegativity," then "the terms corresponding to A^+B^- and A^-B^+ will occur with the same coefficient"; that is, $\alpha = \beta$. For a highly polar molecule with A positive, α exceeds γ and β .

Starting with the above idea, Pauling has succeeded in setting up an empirical scale of approximate relative electronegativities on the basis of thermal data interpreted with the help of a simple quantum-mechanical argument.² The consistency of observed thermal data with relations demanded by the scale is rather good, indicating that a fairly definite electronegativity value can be assigned to each element, as one would indeed expect from general chemical experience. In connection with these and other electronegativity values, however, it should be borne in mind that we do not know electronegativity to be a concept capable of exact quantitative definition.

We may now inquire whether there is any way of determining theoretically the conditions under which $\alpha = \beta$ is to be expected in Eq. (1), other than when $A \equiv B$. Let us suppose we have a case where the energy required to produce $A^+ + B^-$ from $A + B$ is the same as that required to produce $A^- + B^+$. Further, let us imagine that A^+ and B^- approach each other without deformation, until the distance between their nuclei is equal to the value r_e which occurs in the actual molecule AB; and the same for A^- and B^+ . If the

¹ Cf. J. C. Slater, *Phys. Rev.* **35**, 514-5 (1930) for the case of H_2 . Also S. Weinbaum, *J. Chem. Phys.* **1**, 593 (1933).

² Cf. L. Pauling, *J. Am. Chem. Soc.* **54**, 3570 (1932).

energy given out should be the same in the two cases, then the net energy of the two imagined forms A^+B^- and A^-B^+ would be the same, for $r=r_e$ (and nearly, if not exactly, so for other values of r). Then in the actual molecule, according to quantum mechanics, both should participate equally, i.e., $\alpha=\beta$ in Eq. (1), provided (see later paper in this journal)

$$\begin{aligned} & \int \psi(A-B)H\psi(A^+B^-)d\tau \\ & -E \int \psi(A-B)\psi(A^+B^-)d\tau \\ & = \int \psi(A-B)H\psi(A^-B^+)d\tau \\ & -E \int \psi(A-B)\psi(A^-B^+)d\tau, \end{aligned}$$

a condition whose approximate fulfillment, for two atoms A and B of equal electronegativity, seems plausible (E =actual energy of ψ_{AB}).

Actually, we cannot expect the conditions mentioned, in particular that of equal energy of attraction of A^++B^- and A^-+B^+ , to be met exactly. The equal energy condition would, of course, always be fulfilled if only the simple inverse-square attraction of opposite charges were involved. To this is added, however, a balancing repulsive force which may in general be expected to differ in the two cases. No simple means of estimating the magnitudes of such differences is evident, although crystal lattice data would have some bearing. These are, however, not available for the particular combinations A^+B^- and A^-B^+ in which we are mainly interested.

Let us, nevertheless, as a trial assumption to be tested empirically, suppose that this difference between A^+B^- and A^-B^+ can be neglected, and that the other condition mentioned above is fulfilled. Then whenever we have

$$I_A - E_B = I_B - E_A, \quad (2)$$

we expect $\alpha=\beta$ in Eq. (1), so that A and B are of equal electronegativity according to Pauling's definition. Here I and E stand for ionization potential and electron affinity, respectively; both may conveniently be expressed in volts. If (2) holds, then

$$I_A + E_A = I_B + E_B. \quad (3)$$

[That the equality of the quantities $(I+E)/2$ for two atoms is an approximate criterion for their equal participation in a chemical bond has already been stated by Hund,³ presumably on the basis of an argument like that just given, although Hund does not go through this, nor refer to the concept of electronegativity: cf. subsequent paper in later issue for further details.] According to Eq. (3), two univalent atoms have equal electronegativity if the sum—or average—of ionization potential plus electron affinity is the same for each.

This suggests that we may define the absolute electronegativity of an atom A approximately as $I_A + E_A$ or, probably better for some purposes, as $(I_A + E_A)/2$. Note that I_A and E_A are really the ionization potentials of A and A^- , respectively. The suggested definition carries with it the implication that the electronegativity of an atom is approximately independent of its closeness of approach to another atom. For our definition is based on the assumption that the net total energy, at $r=r_e$ of AB, is nearly the same for A^++B^- as for A^-+B^+ ; it is of course, certain that this equality holds for large values of r (Coulomb forces only).

Referring back to Eq. (1), one can readily see qualitatively why E_A and I_A are of equal importance for the electronegativity of an atom A. The negativeness of atom A in ψ_{AB} is greater the larger the coefficient β and the smaller the coefficient α . (For atom B the relations are reversed.) Coefficient β is large if $\psi(A^-B^+)$ is of low energy, and this is favored by large E_A . Coefficient α is small if $\psi(A^+B^-)$ is of high energy, and this is favored by large I_A . One sees then that large E_A and large I_A both help to promote negativeness of atom A in AB. The fact that precisely the quantity $(E_A + I_A)/2$ is a good measure of electronegativity is less obvious, but will be shown *empirically* in the following, by comparison of an $(E_A + I_A)/2$ scale with other empirically established scales of electronegativity. In a subsequent paper, the theoretical basis of the various electronegativity scales will be gone into more thoroughly.

The foregoing discussion indicates the necessity of considering both the $\psi(A^+B^-)$ and the

³ F. Hund, *Zeits. f. Physik* **73**, 1 (1931); cf. especially pp. 18-19.

TABLE I. *Electroaffinities.*

Atom and state (a)	Abs. electroaff. (b)	Electroaff. rel. to H(b-7.12) (c)	(c) ÷ 2.70 (d)	Thermal (e)	Elec. Mom. × 10 ¹⁸ (f)	Atom and state (a)	Abs. electroaff. (b)	Electroaff. rel. to H(b-7.12) (c)	(c) ÷ 2.70 (d)	Thermal (e)	Elec. Mom. × 10 ¹⁸ (f)
H(s, V ₁)	7.12	0.00	0.00	0.00	0.00	B(s ² p, V ₁)	3.75 ± 0.3	-3.37	-1.25		
F(p ³ , V ₁)	12.66 ± 0.2	5.54	2.05	2.00	—	(s ² p ² , V ₃)	4.21 ± 0.3	-2.91	-1.08		
Cl(p ⁵ , V ₁)	9.81 ± 0.2	2.69	1.00	0.94	1.03		10.28 ± 0.3 _s	3.16	1.17		
Br(p ⁵ , V ₁)	9.06 ± 0.2	1.94	0.72	0.75	0.78		6.23 ± 0.3 _q	-0.89	-0.33		
I(p ⁵ , V ₁)	8.23 ± 0.3	1.11	0.41	0.40	0.38	Be(sp, V ₂)	3.06 ± 0.2	-4.06	-1.51		
O(p ⁴ , V ₂)	10.02 ± 0.5	2.90	1.07	1.40	1.53		6.33 ± 0.2 _s	-0.79	-0.29		
	11.4 ± 1:II	4.3	1.6		(104° angle in H ₂ O)		4.69 ± 0.2 ₇	-2.43	-0.90		
N(s ² p ³ , V ₃)	7.40 ± 0.4	0.28	0.10			Li(s, V ₁)	2.86 ± 0.2 _s	-4.26	-1.58		
	[8.92]:II	[1.8?]	[0.66?]			S				(0.43)	0.68 (if 92° angle in H ₂ S)
(sp ⁴ , V ₃)	8.50 ± 0.4	1.38	0.51	0.95	1.04	P				(0.10)	0.36 (if 98° angle in PH ₃)
	20.76 ± 0.7 _s	13.64	5.05		(if 100° angle in NH ₃)	As				—	0.10 (if 96° angle in AsH ₃)
	12.58 ± 0.5 _q	5.46	2.02								
C(s ² p ² , V ₂)	5.38 ± 0.3	-1.74	-0.64								
(sp ³ , V ₄)	5.93 ± 0.3	-1.19	-0.44	[-0.28]p							
	14.90 ± 0.3 _s	7.78	2.88	[3.04] _s							
	8.17 ± 0.3 _q	1.05	0.39	0.55	(0.5)						

Notes for Table I. In column (b), the electroaffinities are all of the first stage type (one-half the energy difference between a singly-charged positive and negative ion each in the same valence state V_{l-1} where l is as given in column (a)), except those marked II. The latter are second-stage electroaffinities (one-fourth the energy difference between O^{++} , V_0 and O^- , V_0 or N^{++} , V_1 and N^- , V_1). The electron added or removed is a p electron, except in the cases marked s or q . For the s - p hybrid types q , the electroaffinities given are obtained from a weighted mean of s and p values (1 : 3 weight for sp^3 , 1 : 2 for sp^2 and sp^4 , 1 : 1 for sp). This method is not quite right* for q in all cases, but probably nearly enough so, since all values are rather rough. Necessary numerical data were taken from Table IV, except the electron affinity of H (0.715 volt), calculated quantum-mechanically by E. A. Hylleraas (Zeits. f. Physik 65, 209 (1930)). The values in column (e) are from Pauling's paper² (units = volts^{1/2}), except the s and p values for carbon, given in brackets; these have been obtained by applying to the s and p values in column (d) the same correction (0.16) that is needed to make the q value in

column d agree with the thermal (presumably q) value in column (e). It is thought that the column (e) values thus obtained are likely to be more trustworthy than the column (d) values. The values in column (f) are from electric moment data and are in electric moment units, based on compounds H_mX ;⁶ Malone's value of 1.32 for O, which was based on an assumed HOH angle of 90° in H₂O, has here been corrected in accordance with Mecke's band spectrum determination of this angle as 104° 40'. In column (d) the data of column (c) have been divided by 2.70 to assist comparison with columns (e) and (f). Values in parentheses in columns (e) and (f) are uncertain; the value in column (e) for iodine also appears somewhat uncertain. All values for *polyvalent* elements in columns (b), (c) and (d) are distinctly uncertain, mainly because of the uncertainty of the electron affinity data available for these elements; rough estimates of uncertainty are given in column (b). *Added in proof:* If we use (c) ÷ 2.75 for column (d), the values 2.01, 0.98, 0.70 and 0.40 are obtained for F, Cl, Br, I, in close agreement with column (e).

$\psi(A-B^+)$ contributions in building up a good wave function for a moderately polar molecule, e.g., HCl. Even in cases like NaCl, it may not always be wise to neglect the $\psi(Na-Cl^+)$ term.

The "absolute electronegativities" $(I+E)/2$ for H, F, Cl, Br, I, and other atoms are given in Table I. Taking $I+E$ for H, we get 14.25 volts. For the other atoms, it is clear that the ordinary minimum ionization potentials and electron affinities must be replaced by corrected values. This we see on considering Eq. (1) as applied, for example, to HCl. ψ_{AB} for (the normal state of)

HCl is of the $^1\Sigma^+$ type; this is to be built partly from $H+Cl$, partly from H^++Cl^- and from H^-+Cl^+ . For H^++Cl^- , there is no difficulty, Cl^- being in a 1S state which necessarily gives $^1\Sigma^+$ on combination with H^+ . In the case of H^-+Cl^+ , H^- is $1s^2$, 1S , but Cl^+ , if we use its normal state, is s^2p^4 , 3P . Now $^3P(Cl^+)+^1S(H^-)$ can give only triplet states of HCl. [The rule that $^3A+^1B$ cannot give 1AB , but only 3AB , is no longer strict for heavy atoms like I , but nevertheless still holds approximately there in the sense that ψ_{AB} , which is 1AB in the case we are considering, should still contain only rather small contributions from $(^3A^+)+(^1B^-)$.]

* See reference 6.

With $H^-(^1S)$ we must then use a singlet state or states of Cl^+ . There are two such states of low energy, namely, 1D and 1S of s^2p^4 . To calculate the proper I for Cl , we should take a value corresponding to some sort of average of the 1D and 1S of Cl^+ .

The nature of this average can be determined by observing that the four chlorine outer-shell p electrons ($3p$) which are not used for valence purposes in HCl are necessarily in the $3p\pi$ condition ($m_l = \pm 1$). The two electrons which form the bond are in the σ condition ($m_l = 0$). These statements apply to all three of the terms $\psi(H-Cl)$, $\psi(H^+Cl^-)$, $\psi(H^-Cl^+)$ which make the principal contributions to ψ_{HCl} . The electronic structure of HCl may be described as $1s^2 2s^2 2p^6 3s^2 3p\pi^4 \sigma^2$, all but the last two electrons belonging definitely to the Cl atom. In $\psi(H-Cl)$, one σ electron is $3p\sigma$ of Cl , the other is $1s$ of H ; in H^+Cl^- , both σ electrons are $3p\sigma$ of Cl^- ; in H^-Cl^+ , both are $1s$ of H . We are now interested in H^-Cl^+ ; we see that the proper state of Cl^+ for union with H^- is $\dots\dots\dots 3p\pi^4$, which is a singlet state ($^1\Sigma^+$ in fact). Unlike 1D and 1S , however, this is not a real state of uncombined Cl^+ ; nevertheless it has a meaning for *combined* Cl^+ .

It belongs to the category of "valence states" (cf. section C, below). It is a valence state of type $s^2p\pi^4$, V_0 (here V_0 means zero valence in respect to homopolar bonding). Its energy will be found under the heading $s^2\pi^2\pi^2$, V_0 in Table III and s^2p^4 , V_0 in Table IV. This has been estimated with the help of a method of Slater,⁴ and is found to be between the energies of 1D and 1S , as expected. Although there are some difficulties in applying the method (cf. Table III), and although spectroscopic data on the 1D and 1S states are available only for F (not for Cl , Br , I), it has been possible to estimate the desired I values for F , Cl , Br , and I with an error which probably does not exceed ± 0.5 volt at most. Using these corrected I values and the ordinary E values, we get the absolute electronegativities given in Table I. [Actually, I and E values for the valence state s^2p^5 , V_1 of the halogen, given in Table IV, have been used, instead of the slightly

different I and E values for the normal state s^2p^5 , $^2P_{1/2}$; but the sum $I+E$ is the same.]^{4a}

Besides the "absolute electronegativities" just obtained, the *relative* electronegativities referred to hydrogen as zero are given in Table I, and are seen to behave in a reasonable way. It may be noted that if we had used the ordinary ionization potentials of the halogens, this would not be the case; for example, the value for iodine relative to hydrogen would be negative. Comparing the relative electronegativities here obtained with those found by Pauling from thermal data, it is found (cf. Table I, columns d and e) that the two sets of values are proportional within the uncertainties of the thermal-data scale and of the I and E data. A similar agreement is found (column f) with another scale of relative electronegativities, based on electric moments.⁵

The agreements just found, even though probably to some extent fortuitous, indicate that the present "absolute" scale is at least roughly correct for univalent atoms. Granting this, the fact that "absolute" electronegativities are at least roughly proportional to the quantity $(I+E)$ gives new insight into the physical interpretation of the concept of electronegativity.

If the present method of calculating absolute electronegativities is a good one, the scale can readily be extended almost immediately to univalent electropositive atoms (Li , Na , K , $\dots$, Cu^I , Ag , Au^I , perhaps Tl^I , etc.). The necessary I values are known, while the electron affinities are doubtless small enough so that even the uncertain estimates which can be made of their

^{4a} At this point a question which might be raised will be answered. Why cannot we make use of excited triplet states of H^- , e.g., $1s2s$, 3S , in combination with 3P of Cl^+ , since $^3S+^3P$ is capable of giving $^1\Sigma^+$ of HCl ? The energy of $1s2s$, 3S and other triplet states should be only about 0.7 volt higher than that of $1s^2$, 1S , if we assume zero electron affinity for nx , $n > 1$, in H^- , and the total energy of $^3S+^3P$ would be considerably lower than that of $(1s^2, ^1S)$ plus $(3p^4, ^1D$ or $^1S)$. It is true that such terms should contribute to $^1\Sigma^+$ of HCl , but these contributions would probably be small compared with the others, for the reason that Cl^+ (3P) would give rise to an electron configuration $3p\pi^3 3p\sigma$ or $3p\pi^2 3p\sigma^2$, instead of the required $3p\pi^4$. The question just raised and answered for H^- has its analogue for Cl^- and the others, although because of the much larger electron affinity for formation of unexcited Cl^- , etc., the (unstable) excited states are relatively much higher in energy than for H^- , and so need much less to be considered.

⁴ J. C. Slater, Phys. Rev. **34**, 1293 (1929). J. H. Van Vleck, Phys. Rev. **45**, 405 (1934) has recently given a simpler method of getting the energy formulas.

⁵ H. M. Smallwood, Zeits. f. physik. Chemie **B19**, 242 (1932). J. G. Malone, J. Chem. Phys. **1**, 197 (1933); M. G. Malone and A. L. Ferguson, J. Chem. Phys. **2**, 99 (1934).

values should give fairly satisfactory results (cf. Li in Table I, for which $I=5.37$, E (est.) $=0.34$). With this inclusion of electropositive atoms (which give negative electronegativities relative to H), the scale might perhaps better be described as a scale of electroaffinity rather than of electronegativity,—if we may return to a somewhat obsolescent word. Essentially correct results might probably be expected for all univalent s -valent atoms if their electroaffinities are taken proportional to their ionization potentials, with the same proportionality constant as for Li.

B. POLYVALENT ATOMS

We may next consider whether, and, if so, how, the present scale of electroaffinities can be extended to polyvalent atoms. The two other scales (thermal and electric-moment), which have already been developed for a few polyvalent atoms (but only for the case of single bonds), can serve here as an empirical guide and check.

As an example of a polyvalent atom, we may consider oxygen. Instead of Eq. (1) we now have, for a compound AOB, or more simply A_2O , the following (we neglect terms such as $\psi(A^+OA^-)$, $\psi(A-A, O)$, etc.):

$$\begin{aligned}\psi(A_2O) = & a\psi(A-O-A) + b\psi(A-O^+A^+) \\ & + c\psi(A-O^+A^-) + d\psi(A^+O=A^+) \\ & + e\psi(A^-O^{++}A^-). \quad (4)\end{aligned}$$

Suppose, for purposes of argument, that the energy required to produce $A+O^++A^-$ from $A+O+A$ should be the same as to produce $A+O^-+A^+$, and to produce $2A^++O^-$ the same as to produce $2A^-+O^{++}$. [Actually, the co-existence of these two relations is not likely to be fulfilled more than approximately, at best.] Then we should have

$$I_O - E_A = I_A - E_O, \text{ hence } I_O + E_O = I_A + E_A,$$

as in Eqs. (2), (3). Also we should have

$$\begin{aligned}II_O - 2E_A = 2I_A - EE_O; \text{ and} \\ (II_O + EE_O)/2 = I_A + E_A, \quad (5)\end{aligned}$$

where II_O , EE_O are the double ionization potential and electron affinity ($O \rightarrow O^{++}$, $O \rightarrow O^-$). If the net energy of coming together, undeformed, to give $A-O$ distances equal to those in the actual molecule A_2O , should be the same for

AO^+A^- as for AO^-A^+ , and for $A^+O=A^+$ as for $A-O^{++}A^-$, and if also

$$\begin{aligned}\int \psi(A-O-A)H\psi(A-O^+A^-)d\tau \\ \approx \int \psi(A-O-A)H\psi(A-O^-A^+)d\tau,\end{aligned}$$

etc., then we might define $\frac{1}{2}(I_O + E_O)$ and $(II_O + EE_O)/4$, respectively, as the *first-stage* and *second-stage electroaffinities* of the O atom. [Note that the energies of coming together of AO^-A^+ , $A^+O=A^+$, etc., would be more complicated here than in the case earlier considered of a molecule AB; there would be energies of interaction between the two A^+ atoms, or the A and A^+ , for example; hence the likelihood of correctness of the suppositions just made is harder to check than in the case of AB.]—A similar need for considering both first- and second-stage electroaffinities occurs, it may be noted, in double-bonded molecules, e.g. BeO.

The values of I_O and II_O must, of course, be chosen to correspond to the right "valence states" of O^+ and O^{++} , which are $O^+(s^2p^3, V_1)$ and $O^{++}(s^2p^2, V_0)$, respectively. This is because in O^++H^-+H , the O must be in a doublet state (V_1 , intermediate between 2D and 2P of s^2p^3), since only in this way can the spins of O^+ , H^- , and H (respectively, $\frac{1}{2}$, 0, $\frac{1}{2}$) combine to give the zero resultant spin characteristic of H_2O in its normal state. It may be noted that here the O^+ forms one homopolar bond (with the neutral H), one ionic bond with the H^+ . In the case of $O^{++}+H^-+H^-$, the O has to be in a singlet state (V_0 , intermediate between 1D and 1S ; cf. the valence state of Cl^+ in $H-Cl^+$, above), since both H ions are in singlet states (zero spin). The O^- and O^- ions in H^+HO^- and $H^+H^+O^-$ are respectively in the states s^2p^5 , $^2P = V_1$ and s^2p^6 , $^1S = V_0$.

If one knew that b and c in Eq. (4) were much larger than d and e , one might expect the first-stage electronegativity to agree with Pauling's scale value; or if d and e were much larger than c and d , the second-stage electronegativity might be expected to show such an agreement. If b and d should be larger than c and e , it is difficult to say what might be expected. If b, c, d, e are all of since sp^4 has one s and two p valence electrons, it would be possible to get agreement with the

comparable magnitude, some average of first- and second-stage electronegativities might be suitable. The following data suggest that the actual situation may be of this nature.

Table I shows that the estimated first-stage and second-stage electronegativities of O (whose values, it should be noted,—cf. Table I,—are somewhat uncertain, because of uncertainty in the electron affinities) do not differ very greatly, and that their average agrees fairly well with the electronegativities estimated from thermal and electric moment data. This is encouraging.

For nitrogen we ought to consider a first-stage, a second-stage, and a third-stage electronegativity, but it is impossible to estimate the electron affinity for the process $N \rightarrow N^=$ with any accuracy. The value for $N \rightarrow N^-$ is also a mere guess, and that for $N \rightarrow N^{--}$ is decidedly uncertain. According to Table I, both first-stage and second-stage electronegativities (although the value for the latter is doubtful) give values which are definitely too low compared with thermal and electric moment data if, as is usual, we assume the N atom in its ordinary valence state (V_3) to have the normal-state configuration s^2p^3 . For the ions N^+ and N^{++} , as is easily shown, we should then use the valence states s^2p^2 , V_2 , and s^2p , $^2P = V_1$, respectively. The valence state s^2p^2 , V_2 , it may be noted ($s^2p_xp_y$ in Table II), is neither a triplet nor a singlet state, but really a mixture of both, with emphasis on the triplet, however.—It appears, incidentally, that the third-stage electronegativity for s^2p^3 , V_3 , if we may extrapolate from the first- and second-stage values, might agree with the empirical data.

There is, however, another possibility, namely, that we ought to consider not the electronegativity of the s^2p^3 configuration, but something intermediate between this and that of the sp^4 configuration. The sp^4 , like s^2p^3 , gives only three valence bonds, but because of a partial mixing of s with p valence, can give stronger bonds. One can calculate an s as well as a p electronegativity for sp^4 by taking I and E ; the first-stage s and p electronegativities for sp^4 of nitrogen are given in Table I. It will be seen that by taking a suitably weighted mean of first-stage electronegativities for s^2p^3 (here only p valence and electronegativity are possible) and for sp^4 (here we should take a 1 : 2 weighted mean of s and p electronegativities,

empirical values. On calculation, one finds that a 56 percent participation of s^2p^3 and 44 percent participation of sp^4 in the nitrogen valence would give a first-stage electronegativity agreeing with the thermal value 0.95. Since such a relative participation of s^2p^3 and sp^4 appears not unreasonable, we have some support for the attractive possibility that we may get nearly correct absolute electronegativities by using just first-stage values.

This idea gains further support, as we shall see, from the evidence on carbon. In the case of oxygen, it is also not unreasonable. The calculated first-stage electronegativity is there lower than the empirical values, which might be explained by a need of considering the second-stage electronegativity too; but it seems quite possible that a small participation of sp^5 instead of exclusively s^2p^4 , in the bivalence of oxygen, might serve to account for the discrepancy on the basis of first-stage electronegativities alone. Without denying that higher-stage electronegativities may be important and possibly needed for best results,—but noting that, fortunately, they seem not to differ radically from first-stage values,—we shall from now on assume that approximately correct values of the true effective electronegativities of atoms can be obtained by using first-stage electronegativities.

Turning to carbon, we find that in its normal electron configuration s^2p^2 with valence two, this element should be decidedly electropositive, i.e., below hydrogen on the electronegativity scale. Carbon with valence four, however (sp^3 , V_4), according to our method, if we take the first-stage electronegativity averaged over the one s and three p valences, is electronegative relative to hydrogen, the result (0.39 ± 0.1) here obtained being in rather good agreement with the value (0.55) given by Pauling (cf. Table I). The process of averaging, with weight 1 : 3, the s and p electroaffinities of the valence state sp^3 , V_4 should probably be nearly equivalent to calculating the first-stage electronegativity for the Pauling-Slater tetrahedral or q type of orbital for the valence state q^4 , V_4 (cf. Van Vleck⁶

⁶ J. H. Van Vleck, J. Chem. Phys. **2**, 22 (1934). (Van Vleck considers a range of valence states of carbon from q^4 , V_4 (complete s - p hybridization, $\mu = \frac{1}{4}$) to sp^3 , V_4 (no hybridization, $\mu = 1$ in Van Vleck's Eq. (7).)

for a discussion of the valence states sp^3 , V_4 and q^4 , V_4). The good agreement just noted gives added support to the present method, and gives considerable confidence that it may be extended to other cases, a few examples of which are given in Table I.

In connection with the use of first-stage electronegativities for polyvalent atoms, a consideration of diatomic radicals may prove instructive. In the case of the single O—H bond in the radical OH, for example, the proper electronegativity for the O atom is found to be just the first-stage electronegativity, if we apply the reasoning used above for the type AB; the result is unaffected by the potential second valence of O. If in H_2O each of the two single bonds is very similar to the one bond in OH, as is generally supposed, then the first-stage electronegativity of O should be appropriate for O in H_2O as well as in OH. A similar result should hold in other cases. (This would still be true even if the actual A—B bond in a diatomic radical should differ from that in a saturated molecule. In CH, for instance, the actual C—H bond may be formed almost wholly by a carbon p orbital whereas in CH_4 the C—H bonds may be constructed with the help of q carbon orbitals. Nevertheless, each C—H bond in CH_4 might closely resemble the hypothetical case of a C—H single bond formed in C—H by a carbon q electron, justifying the use of a first-stage q electronegativity for carbon in CH_4 .)

A variety of applications of electronegativity values calculated by the present method can be made. Because of the intimate relation of electroaffinity to thermal data and to electric moments, as shown by the setting up of electronegativity scales on the basis of the latter, various predictions in regard to dipole moments, bond moments, and energies of formation might be made from electroaffinities calculated by the present method. For instance, single molecules of LiI, BeO, and HF should have bonds of about equal polarity and also of about equal electric moment, if the parallelism of electric moments to electronegativity differences continues to hold with the more electropositive elements. LiH molecules should be intermediate between HCl and HF in polarity. If compounds with divalent carbon could be isolated, the carbon should be

positive relative to hydrogen; this would probably be expected in the radical CH_2 , whereas in CH_4 the carbon is supposed to be relatively negative.

No attempt will be made here to make extensive predictions; the method by which this can be done for heats of formation of compounds can be seen from Pauling's paper.² Very roughly quantitative predictions could be made by using the electronegativities in column (d) of Table I, which have been reduced to approximately the same numerical values as Pauling's scale. Many more predictions could be made by working out the scale values for all the chemical elements. In many cases, however, this cannot yet be done satisfactorily because of insufficient spectroscopic data. Most of the scale values in Table I can be made more accurate when additional or more reliable spectroscopic data and, in particular, more accurate electron affinities become available. It must also be remembered that the scale is as yet largely empirical in character, and, especially for polyvalent atoms, needs further investigation and testing from both theoretical and experimental standpoints.

A result which should hold quite generally in cases where valence can be varied in steps of two by exciting inner electrons is that, for a given atom, electronegativity increases with increasing valence. In other words, the lowest valence is the most electropositive. This is perhaps nothing new; what is new here is the scale for measuring these differences. For example, trivalent boron is 0.33 unit electropositive relative to hydrogen in Table I, column c , while monovalent boron, if it were stable, would be 1.25 units electropositive. A similar result is to be expected in other cases, e.g., Tl, where both valences are actually realized. Similarly divalent carbon is 0.64 unit electropositive, tetravalent carbon 0.39 unit electronegative; a similar difference is to be expected in Sn, where both valences are easily realized. The difference here results chiefly from the higher electronegativity of s than of corresponding p valences. In more complex atoms, d valences also enter. A calculation of electroaffinities for the various valence states of Cu, Ag, Au, Zn, Cd, Hg would be interesting, although spectroscopic data are somewhat incomplete.

TABLE II. Interaction energies for atomic states belonging to configurations $1s^2 2s^m 2p^n$ (cf. also Table IIA).

State	G_1	G_2	State	G_1	G_2	State	G_1	G_2
$s^2, {}^1S = V_0$	0	0	sp^3 See Table IIA			$(s^2)p^4, {}^3P$	-4	-15
$sp, {}^3P$	-1	0	$(s^2)p^3, {}^4S$	-3	-15	1D	"	-9
1P	1	0	2D	"	-6	1S	"	0
V_2	$-\frac{1}{2}$	0	2P	"	0	x^2yz, V_2	"	$-13\frac{1}{2}$
$(s^2)p, {}^2P = V_1$	-1	0	xyz, V_3	"	$-10\frac{1}{2}$	$\pi^2\pi\sigma, V_2$	"	$-13\frac{1}{2}$
$sp^2, {}^4P$	-2	-5	$\pi\pi\sigma, V_3$	"	-9	$\sigma^2\pi\pi, V_2$	"	-12
2P	1	-5	$x^2y, \sigma^2\pi, \}$	"	-3	$\pi^2\pi^2, x^2y^2, V_0$	"	-6
2D	-1	1	$\pi^2\pi, V_1$	"	-6	$\sigma^2\pi^2, V_0$	"	-9
2S	-1	10	$\pi^2\sigma, V_1$	"	-6			
$sxy, s\pi\sigma, V_3$	-1	$-3\frac{1}{2}$	$sp^4, {}^4P$	-3	-15	$sp^5, {}^3P$	-3	-20
$s\pi\pi, V_3$	-1	-2	2P	0	-15	1P	-1	"
$sx^2, s\sigma^2, V_1$	-1	4	2D	-2	-9	sx^2y^2z, V_2		
$s\pi^2, V_1$	-1	1	2S	"	0	$s\pi^2\pi^2\sigma, V_2$	$-2\frac{1}{2}$	"
			sx^2yz, V_3	"	$-13\frac{1}{2}$	$s\pi^2\pi\sigma^2, V_2$		
$(s^2)p^2, {}^3P$	-2	-5	$s\pi^2\pi\sigma, V_3$	"	$-13\frac{1}{2}$	$(s^2)p^5, {}^2P = V_1$	-5	-20
1D	"	1	$s\sigma^2\pi\pi, V_3$	"	-12	$sp^6, {}^2S = V_1$	-3	-30
1S	"	10	$s\pi^2\pi^2, \}$	"	-6	$(s^2)p^6, {}^1S = V_0$	-6	-30
$xy, \pi\sigma, V_2$	"	$-3\frac{1}{2}$	sx^2y^2, V_1	"	-6			
$\pi\pi, V_2$	"	-2	$s\sigma^2\pi^2, V_1$	"	-9			
x^2, σ^2, V_0	"	4						
π^2, V_0	"	1						

Explanation of Table II: (a) The symbols $s, p, x, y, z, \pi, \pi\pi, \pi^2, \sigma$, here respectively denote $2s, 2p, 2p_x, 2p_y, 2p_z, 2p\pi(2p_{+1} \text{ or } 2p_{-1}), 2p\pi_+2p\pi_-, (2p\pi_+)^2 \text{ or } (2p\pi_-)^2, 2p\sigma(2p_0)$, while (s^2) indicates either $(2s)^2$ or nothing. The letter V denotes a "valence state," whose valence is given by the subscript. G_1 means $(1/3)G^1(2s, 2p)$, G_2 means $(1/25)G^2(2p, 2p)$ of Slater.⁴ The numbers in the columns headed G_1 and G_2 give the numerical factor with which G_1 or G_2 appears in the energy expression for the state in question.

(b) Certain terms which are constant for any one electron configuration have been omitted from Tables II and IIA. The omitted terms which come from integrals

of the types which Slater calls J 's and K 's are given by the following formula: $F^0(1s, 1s) + \alpha F^0(2s, 2s) + 2mF^0(1s, 2s) + 2nF^0(1s, 2p) + mnF^0(2s, 2p) - mG^0(1s, 2s) - (n/3)G^1(1s, 2p) + \beta G^0(2p, 2p)$, where $\alpha=1$ if $m=2$, otherwise $\alpha=0$, and where $\beta=0, 0, 1, 3, 6, 10, 15$, for $n=0, 1, 2, 3, 4, 5, 6$, respectively.

(c) The way in which β , and the coefficients of G_1 and G_2 in Table II, were obtained is illustrated by Table IIA. The remaining F 's and G 's in the formula just given under (b) correspond to J 's and K 's involving pairs of electrons not both of $2p$ type; in connection with them cf. Slater's tables, reference 4, p. 1312.

C. APPENDIX: INTERACTION INTEGRALS AND ENERGY DATA FOR CONFIGURATIONS $1s^2 2s^m 2p^n$

In the following Table II are given the numerical factors with which certain interaction integrals called J 's and K 's, or F 's and G 's, occur in Slater's method⁴ of approximating the energies of the states of complex atoms. The total energy of an atom in a given state is approximately equal to the sum of certain terms which are constant for a given electron configuration, plus others which vary from one state to another of the configuration.

In addition to the actual atomic states associated with the various electron configurations in Table II, whose energies have already been given

by others,^{6a} energy expressions are also tabulated for certain "valence states." The term valence state, introduced by Van Vleck,⁶ describes a certain hypothetical state of interaction of the electrons of an atomic electron configuration. In formulating these, we shall here most often use the classification $2p_x, 2p_y, 2p_z$ for the $2p$ orbitals

$$(2p_z \equiv 2p_0, 2p_x = 2^{-1/2}(2p_{+1} + 2p_{-1}),$$

$$2p_y = 2^{-1/2}i^{-1}(2p_{+1} - 2p_{-1})).$$

Still other types of valence states, involving various hybrid orbitals formed by mixing $2s$ and $2p$, could be given, but are omitted from the

^{6a} Slater, Condon and Shortley, in part, and recently Van Vleck (Reference 4, Eqs. 22, 38).

TABLE IIA. Illustration of derivation of results in Table II.

State	$J_{\sigma\sigma}$	$J_{\sigma\pi}$	$J_{\pi\pi}$	$K_{\sigma\pi}$	$K_{\pi\pi}$	G_1	$G^0(2p, 2p)$	G_2
$sp^3, {}^5S$	0	2	1	-2	-1	-3	3	-15
3S	-1	4	0	-1	0	1	"	-15
3D	0	2	1	-1	0	-2	"	-6
1D	0	2	1	-1	0	0	"	-6
3P	1	0	2	0	-1	-2	"	0
1P	1	0	2	0	-1	0	"	0
sxy^2, V_4	0	3	0	$-1\frac{1}{2}$	0	$-1\frac{1}{2}$	"	$-10\frac{1}{2}$
$s\pi\pi\sigma, V_4$	0	2	1	-1	$-\frac{1}{2}$	$-1\frac{1}{2}$	"	-9
$sx^2y, s\sigma^2\pi,$ $s\pi^2\pi, V_2$	1	2	0	-1	0	$-1\frac{1}{2}$	"	-3
$s\pi^2\sigma, V_2$	0	2	1	-1	0	$-1\frac{1}{2}$	"	-6

Explanation: Below each heading $J_{\sigma\sigma}$, etc., is given the factor by which this integral is to be multiplied in getting the energy of the state in question; these numbers are determined by Slater's method,⁴ modified in the case of the *valence* (unpaired) electrons in *valence states* in that $-\frac{1}{2}K$ is put down for the K interaction of each two valence electrons (see text). The symbols $J_{\sigma\sigma}$, $J_{\sigma\pi}$, $J_{\pi\pi}$, $K_{\sigma\pi}$, $K_{\pi\pi}$, and $G_1 \equiv (1/3)G^1(2s, 2p)$ refer, respectively, to Slater's $J(2p_0, 2p_0)$, $J(2p_{\pm 1}, 2p_0)$, $J(2p_{\pm 1}, 2p_{\pm 1})$, $K(2p_{\pm 1}, 2p_0)$, $K(2p_{\pm 1}, 2p_{\pm 1})$, and $K(2s, 2p_0, \pm 1)$, and respectively reduce (see Slater's tables,⁴ p. 1312) to $F^0(2p, 2p) + (4/25)F^2(2p, 2p)$; $F^0(2p, 2p) - (2/25)F^2(2p, 2p)$; $F^0(2p, 2p) + (1/25)F^2(2p, 2p)$; $(3/25)G^2(2p, 2p)$; $(6/25)G^2(2p, 2p)$; $(1/3)G^1(2s, 2p)$, with $F^0(2p, 2p) = G^0(2p, 2p)$ and $F^2(2p, 2p) = G^2(2p, 2p)$ here (equivalent p electrons). By using these relations for the J 's and K 's, the results in the last three columns are obtained. The results in Table II were obtained in the same way. A simpler method for the ordinary atomic states has recently been given by Van Vleck (reference 4, Eqs. (22) and (38)). Table IIA gives only those J 's and K 's (including $G_1 = K(2s, 2p)$) which vary for different states of sp^3 .

tables below; such states, often important for atoms having both s and p valences, are discussed by Van Vleck.⁶

It is evident that

$$J(2p_z, 2p_z) = J(2p_y, 2p_y) = J(2p_x, 2p_x),$$

since in the x, y, z classification $2p_x, 2p_y, 2p_z$ are equivalent; these integrals may be denoted $J_{\sigma\sigma}$. It is then easily shown^{6b} that $J(2p_x, 2p_y) = J(2p_x, 2p_z) = J(2p_y, 2p_z) = J(2p_0, 2p_{+1})$, which may be denoted $J_{\pi\sigma}$; and that $K(2p_x, 2p_y) = K(2p_x, 2p_z) = K(2p_y, 2p_z) = K(2p_0, 2p_{+1})$, which may be denoted $K_{\pi\sigma}$. Hence the integrals for electron configurations expressed in terms of $2p_x, 2p_y, 2p_z$ can be tabulated under the headings $J_{\sigma\sigma}$, $J_{\pi\sigma}$, $K_{\pi\sigma}$ in Table II (cf. Table IIA).

^{6b} Cf. Slater, Reference 4, p. 1133.

A "valence state" is an atom state chosen so as to have as nearly as possible the same condition of interaction of the atom's electrons *with one another* as when the atom is part of a molecule. The electrons in question may be classified as non-valence electrons and valence electrons. The non-valence electrons are usually in pairs (e.g., $2p_x^2$ in the configuration $1s^2 2s^2 2p_x^2 2p_y 2p_z$), in which case the m_s values of the two electrons are necessarily opposite in sign so that (cf. Slater) the coefficient of the integral $K(2p_x, 2p_x)$ is zero.

A valence state of an atom is one in which the latter's valence electrons behave *toward one another* as if each were paired somehow with a valence electron of a foreign atom, but not with any valence electron of the given atom. Before going further, we note that this question of pairing of the electrons in a valence state of an atom affects only the integrals K , and then only of electrons not intra-atomically paired. The J 's, for a given detailed electron configuration specified in terms of m_l values or of the x, y, z classification (e.g., $2p_x^2 2p_y 2p_z$ or $2p_{+1} 2p_{-1} (2p_0)^2$), do not depend on the m_s values of the electrons.

The correct contribution of the K 's to the energy for a valence state of a given atom is obtained by writing $-\frac{1}{2}K(i, j)$ for every combination (i, j) of the valence electrons of the given atom taken two at a time; this result, or essentially this, has been obtained by Slater and others.^{6, 7} Any *unpaired* non-valence electron which may be present can be treated for this purpose as a valence electron; this case of unpaired non-valence electrons is of interest when we consider certain ionization processes, e.g., removal of a non-valence electron from a fluorine atom which belongs to an F_2 molecule.

The result just stated is based on the fact that the K interaction between two electrons i, j each in a definite orbital state (definite n, l , and m_l or n, l , and x, y, z , classification) is zero if the two electrons have opposite m_s values ($+\frac{1}{2}, -\frac{1}{2}$), but equal to $-K(i, j)$ if both have the same m_s value. Now any unpaired electron in a valence state (including unpaired non-valence electrons) is in a condition such that the probability that its

⁷ Cf. J. C. Slater, Phys. Rev. **38**, 1109 (1931), discussion of several electrons with only spin degeneracy. The results given there can of course be used even if two or more electrons are on the same atom.

TABLE III. Approximate energies of valence states in volts.

State	Atom or ion; normal state	Energy (above normal state energy)	Method of estimation [N. B. In many cases averages were used (cf. e.g. s^2xy, V_2) because Slater's relations (Table II) do not hold well]	State	Atom or ion; normal state	Energy (above normal state energy)	Method of estimation [N. B. In many cases averages were used (cf. e.g. s^2xy, V_2) because Slater's relations (Table II) do not hold well]
s^2, V_0	$s^2, 1S$	All 0.00		s^2x^2y, V_1 $=s^2\sigma^2\pi, V_1$ $=s^2\pi^2\pi, V_1$	$s^2p^2, 4S$: C- N O+ F++	[1.83] 3.06 4.29 5.45	Extr. $s^2p^2, 2D$ plus aver. of $\frac{1}{2}(2P-2D)$ and $(2D-4S)/3$ Ditto
sp, V_2	$s^2, 1S$: Li- Be B+ C++	[0.97] 3.35 5.70 8.00	Extrapolated from Be, B+ $s^2, 2P$ plus $\frac{1}{2}(P-3P)/4$ Ditto	$s^2\pi\sigma, V_3$	$s^2p^2, 4S$: N O+	1.69 2.35	$s^2p^2, 2D$ minus aver. of $(2D-4S)/3$ and $\frac{1}{2}(2P-2D)$ Ditto
xy, V_2	$s^2, 1S$: Li- Be B+	[2.52] 7.50 12.43	Extr. from Be, B+ $p^2, 2P$ plus $\frac{1}{2}G_2$ (est. $6G_2=6000$) Same method (est. $6G_2=8560$)	$\pi^2\sigma, V_1$	O+	3.32	Same as $2D$ of s^2p^2
z^2, V_0	$s^2, 1S$: Li- Be B+	[3.07] (8.43) (13.75)	Extr. from Be, B+ xy, V_2 plus $\frac{1}{2}G_2$ (see xy, V_2) Ditto	sx^2yz, V_3 $=s\pi^2\pi\sigma, V_3$	$s^2p^2, 4S$: C- N O+ F++	[8.84] (13.86) 18.88 23.84	Extr. $sp^4, 2D-4\frac{1}{2}G_2$, where $\frac{1}{2}G_2$ =(8922) est. from O+, F++; and $2D$ est. from O+, F++; Ditto, with $\frac{1}{2}G_2=(13,113)$ = aver. of $\frac{1}{2}(\frac{1}{2}(2P-4P)-3(2P-2D)/4)$ and $\frac{1}{2}(2S-2D)$ Ditto, with $\frac{1}{2}G_2=(17,147)$ from $4P, 2P, 2D$ and O+
s^2p, V_1	$s^2p, 2P$	All 0.00		sx^2y^2, V_1 $=s\pi^2\pi^2\sigma, V_1$	$s^2p^2, 4S$: C- N O+ F++	[9.81] (15.69) 21.57 27.37	Extr. $sp^4, 2D+3G_2$, with G_2 as for sx^2yz, V_3 Ditto
sxy, V_3	$s^2p, 2P$: Be- B C+ N++ O+++	[2.55] 5.35 8.15 10.88 13.63	Extr. $2D-4\frac{1}{2}G_2$; $9G_2=(8950)$ by extr. from C+ etc. $2D-4\frac{1}{2}G_2$; $9G_2=17,780$, ave. value from $4P, 2P, 2S$ Same method; $9G_2=25,790$ Same method; $9G_2=32,990$	x^2y^2z, V_1 $=\pi^2\pi^2\sigma, V_1$	$s^2p^2, 4S$: C- N O+ F++	[9.81] (15.69) 21.57 27.37	Extr. $sp^4, 2D+3G_2$, with G_2 as for sx^2yz, V_3 Ditto
sx^2, V_1	$s^2p, 2P$: Be- B C+ N++ O+++	[2.60] 6.28 9.97 13.52 17.02	Extr. $2D+3G_2$; G_2 as for sxy, V_3 Ditto Ditto Ditto	s^2x^2yz, V_2 $=s\pi^2\pi^2\sigma, V_2$	$s^2p^2, 3P_2$: N- (see note O b) F+ Cl+ Br+ I+	[0.38] 0.67 0.95 (0.74) (0.78) (0.92)	Extr. Aver. of $(1D-3P)/4$ and $(1D-2P)$ $-\frac{1}{2}(1S-1D)$ Ditto From $(s^2xyz, V_2-s^2p^2, 3P_2)$ $=(V_2-3P_{center\ of\ gravity})+$ $(3P_{c.g.}-3P_2)$, with $3P_{c.g.}$ $-3P_2=(3P_2-3P_0)/3=0.04$, (0.17), (0.35) for Cl+, Br+, I+; and $(V_2-3P_{c.g.})$ est. roughly by assuming ratio of this quantity to ionization poten- tial of atom is same as for F: e.g., for Cl+, est. $(V_2-3P_{c.g.})$ $=0.70=(0.93/17.32)(12.96)$
xyz, V_3	$s^2p, 2P$: C+	(18.0)	$p^2, 4S+\frac{1}{2}(2D-4S)$	$s^2x^2y^2, V_0$ $=s\pi^2\pi^2\sigma, V_0$	$s^2p^2, 3P_2$: O F+ Cl+ Br+ I+	2.81 3.87 (2.92) (2.79) (2.69)	$s^2p^2, 1D$ plus aver. of $\frac{1}{2}(1D-3P)$ and $(1S-1D)/3$ Ditto Rough estimates for Cl+, Br+, I+ made as for V_2 above
z^2y, V_1	$s^2p^2, 3P_0$: B- (see note b) N+ O++ F+++	[0.28] 0.49 0.66 0.89 (1.07)	Extr. Aver. of $(1D-3P)/4$ and $(1D-3P)/4-\frac{1}{2}(1S-1D)$ Ditto Ditto Ditto	sx^2y^2, V_2 $=s\pi^2\pi^2\sigma, V_2$	$s^2p^2, 3P_2$: N- O F+ Ne++	[11.54] 17.08 22.70 28.03	Extr. $sp^5, 3P$ plus est. $(3P-1P)/4$ (est. $=15,120$) $sp^5, 3P$ plus $(3P-1P)/4$ $(3P-1P=75,574)$ $sp^5, 3P$ plus est. $(3P-1P)/4$ (est. =22,470)
s^2x^2, V_0	$s^2p^2, 3P_0$: B- C N+ O++ F+++	[0.90] 1.80 2.73 3.60 (4.43)	Extr. $s^2p^2, 1D$ plus aver. of $(1S-1D)/3$ and $\frac{1}{2}(1D-3P)$ Ditto Ditto Ditto	$x^2y^2z^2, V_0$ $=\pi^2\pi^2\sigma^2, V_0$	$s^2p^2, 2P_{1/2}$: F Cl Br I	0.02 0.04 0.15 0.31	$(2P_1-2P_{1/2})/3$
$sxyz, V_4$	$s^2p^2, 3P_0$: B- C N+ O++ F+++	[4.66] 8.16 11.59 15.09 18.51	Extr. $3D+\frac{1}{2}G_1-4\frac{1}{2}G_2$; $\frac{1}{2}G_1=[10,460]$; $6G_2=(11,200)$ from $s^2D, 3P$ and extr. $3D+\frac{1}{2}G_1-4\frac{1}{2}G_2$; $\frac{1}{2}G_1=13,644$ and $6G_2=16,040$; ave. values from $s^2S, 3D, 3P, 3S, 1D, 1P$ Same method; $\frac{1}{2}G_1=16,973$ and $6G_2=20,055$ $3D+\frac{1}{2}G_1-4\frac{1}{2}G_2$; $\frac{1}{2}G_1=20,448$ and $6G_2=24,610$ from $s^2S, 3D, 3P$, $1D, 1P$ and Extr.	$s^2x^2y^2, V_1=s^2P$ $=s\pi^2\pi^2\sigma, V_1=2P$	$s^2p^2, 2P_{1/2}$: F Cl Br I	0.02 0.04 0.15 0.31	$(2P_1-2P_{1/2})/3$
sx^2y, V_2 $=s\sigma^2\pi, V_2$ $=s\pi^2\pi, V_2$	$s^2p^2, 3P_0$: B- C N+ O++ F+++	[5.68] 9.89 14.06 18.19 22.30	Extr. $3D+\frac{1}{2}G_1+3G_2$; G_1 and G_2 as for $sxyz, V_4$ Ditto Ditto Ditto	$s^2x^2y^2, V_1=s^2P$ $=s\pi^2\pi^2\sigma, V_1=2P$	$s^2p^2, 2P_{1/2}$: F Cl Br I	0.02 0.04 0.15 0.31	$(2P_1-2P_{1/2})/3$
z^2yz, V_2	$s^2p^2, 3P_0$: N+	[27.4]	Rough estimate	$s^2x^2y^2, V_1=s^2S$ $=s\pi^2\pi^2\sigma^2, V_1=2S$	$s^2p^2, 2P_{1/2}$: F Cl Br I	0.02 0.04 0.15 0.31	$(2P_1-2P_{1/2})/3$
z^2y^2, V_0	$s^2p^2, 4S$: C- N O+ F++	[0.79] 1.33 1.85 2.35	Extr. Aver. of $\frac{1}{2}(2D-4S)$ and $(2D-4S)$ $-3(2P-2D)/4$ Ditto Ditto				

Notes: (a) In the above table, values are given for most but not all cases where the necessary data on atomic energies are available either directly or with the help of fairly reliable interpolations or extrapolations. Data used in the calculations were taken from R. F. Bacher and S. Goudsmit, *Atomic Energy States*; additional data on C+, N+, N++, O+, O++, and on F+, F++, F+++ from B. Edlén,

Zeits. f. Physik **84**, 746 (1933) and I. S. Bowen, Phys. Rev. **45**, 82 (1934).

(b) For $s^2p^4, 3P$, corrections of $\frac{1}{2}(3P_2-3P_1)$ have been included to take care of the fact that the component $3P_2$ is below the center of gravity of $3P$ by that amount. For $s^2p^2, 3P$, a correction of $2(3P_2-3P_0)/3$ has been applied to correct for the difference between $3P_{center\ of\ gravity}$ and $3P_0$.

TABLE IV. *Some ionization potentials and electron affinities (volts).* (Most values are only approximate. Valence states given here are only *xyz* states: cf. Table III; ionization potentials and electron affinities for removal of electrons from valence states having σ , π quantization can easily be obtained, when needed, with the help of Tables II, III. Removal of *s* electron is indicated by *s* in Ioniz. pot. column; in all other cases a *p* electron is removed.)

State of atom (or neg. ion)	State of pos. ion (or atom)	Ioniz. pot. (or el. aff.)	State of atom (or neg. ion)	State of pos. ion (or atom)	Ioniz. pot. (or el. aff.)
Li-($s^2, {}^1S = V_0$) Li($s, {}^2S = V_1$)	Li($s, {}^2S = V_1$) Li+(${}^1S = V_0$)	0.34† _s 5.37 _s	N-(s^2p^2, V_1) N(s^2p^2, V_2)	N(s^2p^3, V_2) N+(s^2p, V_1)	[-??] p^2 42.62* p^2
Be-($s^2p, {}^2P = V_1$) (s^2p, V_1) (s^2p, V_1) Be($s^2, {}^1S$) (s, V_2) (s, V_2) (p^2, V_2)	Be($s^2, {}^1S$) (s, V_2) (s, V_2) Be+($s, {}^2S = V_1$) (s, V_1) ($p, {}^2P = V_1$) (p, V_1)	-0.57† 2.78* _s 0.18* 9.28 _s 5.93* 9.87* _s 5.72*	O-($s^2p^2, {}^2P = V_1$) (s^2p^2, V_1) O($s^2p^4, {}^3P_2$) (s^2p^4, V_2) "	O($s^2p^4, {}^3P_2$) (s^2p^4, V_2) O+($s^2p^3, {}^4S$) (s^2p^3, V_2) (s^2p^3, V_1) (s^2p^4, V_3)	2.2 ^a 2.87* 13.55 14.73* 17.17* 31.76* _s
B-($s^2p^2, {}^3P$) (s^2p^2, V_0) (s^2p^2, V_2) (s^2p^2, V_2) B($s^2p, {}^2P = V_1$) (s^2p, V_2) (s^2p, V_2)	B($s^2p, {}^2P = V_1$) (s^2p, V_1) (s^2p^2, V_3) (s^2p^2, V_3) B+($s^2, {}^1S = V_0$) (s, V_2) (p^2, V_2)	0.12† -0.78* 5.19* _s -0.21* 8.28 8.63* 15.36* _s	O-($s^2p^2, {}^1S = V_0$) O($s^2p^4, {}^3P_2$) (s^2p^4, V_2)	O($s^2p^4, {}^3P_2$) (s^2p^4, V_2) O+($s^2p^3, {}^3P_2$) (s^2p^3, V_0)	[-6.5] aa, p^2 [-5.8*] p^2 48.48* p^2 51.41* p^2
F-($s^2p^2, {}^2P = V_1$) (s^2p^2, V_2) (s^2p^2, V_2) F($s^2p^5, {}^2P = V_1$) (s^2p^5, V_2)	F($s^2p^5, {}^2P = V_1$) (s^2p^5, V_2) (s^2p^5, V_2) F+($s^2p^4, {}^3P_2$) (s^2p^4, V_2) (s^2p^4, V_0)	4.13 ^b 4.15 17.32 ^c 18.25* 21.17*	F-($s^2p^2, {}^1S = V_0$) F($s^2p^5, {}^2P = V_1$) (s^2p^5, V_2)	F($s^2p^5, {}^2P = V_1$) (s^2p^5, V_2) F+($s^2p^4, {}^3P_2$) (s^2p^4, V_0)	4.13 ^b 4.15 17.32 ^c 18.25* 21.17*
C-($s^2p^3, {}^4S$) (s^2p^3, V_1) (s^2p^3, V_3) (s^2p^4, V_3) C($s^2p^2, {}^3P$) (s^2p^2, V_2) (s^2p^2, V_2) (s^2p^3, V_4) (s^2p^3, V_4)	C($s^2p^3, {}^3P$) (s^2p^3, V_2) (s^2p^3, V_4) (s^2p^3, V_4) C+($s^2p, {}^2P = V_1$) (s^2p, V_1) (s^2p^2, V_3) (s^2p^2, V_3) (p^2, V_2)	1.37† 0.03* 8.74* _s 0.69* 11.22 10.73* 18.84* _s 11.17* (21.06)* _s	Cl-(s^2p^4, V_0) Cl($s^2p^5, {}^2P_{1/2}$) (s^2p^5, V_1)	Cl($s^2p^5, {}^2P_{1/2}$) (s^2p^5, V_1) Cl+($s^2p^4, {}^3P_2$) (s^2p^4, V_2) (s^2p^4, V_0)	3.75 ^b 3.79* 12.96 13.66* ^{ad} 15.84* ^{ad}
Br-($s^2p^4, {}^3P$) (s^2p^4, V_2) (s^2p^4, V_2) (s^2p^5, V_2) N($s^2p^3, {}^4S$) (s^2p^3, V_2) (s^2p^3, V_2) (s^2p^4, V_3) (s^2p^4, V_3) (s^2p^4, V_3)	N($s^2p^3, {}^4S$) (s^2p^3, V_2) (s^2p^4, V_3) (s^2p^4, V_3) N+($s^2p^2, {}^3P_2$) (s^2p^2, V_2) (s^2p^3, V_4) (s^2p^3, V_4) (s^2p^3, V_2) (p^4, V_2)	0.04† 0.99* 13.52* _s 2.36* 14.48 13.81* 24.77* _s 12.24* 14.63* [28.0*] _s	Br-($s^2p^4, {}^3P$) (s^2p^4, V_2) (s^2p^4, V_2) Br($s^2p^5, {}^2P_{1/2}$) (s^2p^5, V_1)	Br($s^2p^5, {}^2P_{1/2}$) (s^2p^5, V_1) Br+($s^2p^4, {}^3P_2$) (s^2p^4, V_2) (s^2p^4, V_0)	3.53 ^b 3.68* 11.80 12.43* ^{ad} 14.44* ^{ad}
I-(s^2p^5, V_0) I($s^2p^5, {}^2P_{1/2}$) (s^2p^5, V_1)	I($s^2p^5, {}^2P_{1/2}$) (s^2p^5, V_1) I+($s^2p^4, {}^3P_2$) (s^2p^4, V_2) (s^2p^4, V_0)	3.22 ^b 3.53* 10.55 ^c 11.16* ^{ad} 12.93* ^{ad}	I-(s^2p^5, V_0) I($s^2p^5, {}^2P_{1/2}$) (s^2p^5, V_1)	I($s^2p^5, {}^2P_{1/2}$) (s^2p^5, V_1) I+($s^2p^4, {}^3P_2$) (s^2p^4, V_2) (s^2p^4, V_0)	3.22 ^b 3.53* 10.55 ^c 11.16* ^{ad} 12.93* ^{ad}

Notes for Table IV. † indicates electron affinities taken from a paper by Glockler.⁸ These are based on a method of extrapolation similar to one used by Bacher and Goudsmit (private communication), who consider the latter as reliable as any available. They point out, however, that the method implies a power series approximation for ionization energy, which should become less convergent for negative ions (electron affinities). Unfortunately the available accurate electron affinity data are inadequate to serve as a check on the accuracy of the above extrapolation: for H, the extrapolated value agrees closely with the accurate quantum-mechanically calculated value (0.715 volt); for O and F the extrapolated values are about 1 volt lower than the empirical, but the empirical value for O is uncertain, while the data used in the extrapolation are uncertain in both cases. [In the preceding sentence, we are not using Glockler's extrapolated value +3.80 volts for the electron affinity of O, but a value 1.24 volts. Glockler's value is based on the dubious value 18.6 volts for the ionization potential of F, while the value 1.24 is obtained by Glockler's method if the value 17.32 volts estimated by the writer (cf. note *c* below) is used for the ion. pot. of F.] It seems probable that the extrapolated electron affinities of Glockler (except for O and perhaps a few other cases where reliable data were not available) are not in error by amounts greater than ± 0.3 to ± 1.0 volts.

* indicates *valence state* electron affinities and ionization potentials which have been obtained by combining *ordinary*

electron affinities (†, *a*, *aa*, *b*) or *ordinary* ionization potentials (cf. note *c* below) with valence state energy data given in Table III. In the case of the electron affinities, it should be noted that the data (all given in brackets [] in Table III) were obtained by simple extrapolations having a similar basis to that used by Glockler for the *ordinary* electron affinities. In case better values for the *ordinary* electron affinities should be determined in future, the resulting correction needed for the *ordinary* electron affinity of any given atom in Table III should also be approximately right for all the valence state electron affinities of the same atom.

^a is based on a fairly reasonable experimental value 2.2 ± 0.2 volts obtained by Lozier by electron impact methods.⁹ This is probably preferable to the extrapolated value (cf. bracketed comment in † note above).

^{aa} indicates a very rough electron affinity value (150 ± 50 kcal.) based on crystal lattice constants.¹⁰

^b indicates values based on crystal lattice data, and considered accurate to ± 2 percent, given by Mayer and Helmholtz.¹¹ It should, however, be noted that the value for F may be less accurate, because it involves the heat of dissociation *D* of F₂; if for instance the accepted value of

⁸ G. Glockler, Phys. Rev. **46**, 111 (1934).

⁹ W. W. Lozier, Phys. Rev. **46**, 268 (1934).

¹⁰ J. E. Mayer and Maltbie, Zeits. f. Physik **75**, 748 (1932).

¹¹ J. E. Mayer and L. Helmholtz, Zeits. f. Physik **75**, 19 (1932).

D should be raised, the M. and H. value of the electron affinity of F would be raised by $\frac{1}{2}D$. The accepted value of D is 2.80 ± 0.03 volts, but in the writer's opinion the indicated probable error ± 0.03 is far too small, and the value 2.80 itself has no real foundation, although it may happen to be nearly correct. The value 2.80 volts (63.3 ± 0.07 kcal.) is based on a determination of the maximum (λ_{2900}) of the ultraviolet continuous absorption of F_2 by von Wartenberg and Taylor, combined with an extrapolation of the interval $\lambda_{\text{conv.}} - \lambda_{\text{max}}$ as observed in I_2 , Br_2 and Cl_2 , to F_2 ($\lambda_{\text{conv.}}$ = wave-length of convergence of vibrational levels, corresponding to dissociation; not observed in F_2).¹² The extrapolation used (linear variation of $\lambda_c - \lambda_m$ with atomic weight A) has no theoretical basis, and might well be deceptive, since F occupies an exceptional position among the halogens; furthermore, the supposed linear variation of $\lambda_c - \lambda_m$ is actually not even empirically fulfilled for I_2 , Br_2 , Cl_2 when the most reliable

¹² H. von Wartenberg, G. Sprenger and J. Taylor, *Bodenstein-Festband*, p. 61. (*Ergänzungsband* of the *Zeits. f. physik. Chemie*, Leipzig, 1931.)

available data on λ_c and λ_m are used (cf. e.g., Mulliken, *Phys. Rev.* **46**, 549, 1934, Table IIB).

^c indicates for fluorine an estimate of the ionization potential made from data on the s^2p^43p , 4P , 2P states of fluorine, using for the $3p$ term value an estimate based on interpolation in the series N, O, Ne, Na; the present estimate of the ionization potential is believed to be much better than Dingle's value 18.6 volts, which is obviously too large. The estimated value 10.55 volts for iodine is based on interpolation and other comparisons between data on other atoms and is believed to be at least as well founded as other estimates in the literature. The ionization potentials given for all other atoms are spectroscopic values (cf. Bacher and Goudsmit).

^d indicates values for Cl^+ , Br^+ , I^+ which are based on a method of estimation (see Table III) which may give appreciable errors (probably not worse than ± 0.5 , however). If we may judge by a comparison of F^+ , Cl^+ , Br^+ , I^+ with C, Si, Ge, Sn, the values 2.92, 2.79, 2.69 for $s^2x^2y^2$, V_0 in Table III, and so the ionization potentials $V_1 \rightarrow V_0$ in Table IV, are about 0.35 volt too high for Cl^+ , Br^+ , about 0.4 too low for I^+ ; but this is uncertain.

m_s is parallel to that of any other electron in a similar condition in the same atom is just $\frac{1}{2}$. The quantum-mechanical energy integral taken over a state in which parallel and antiparallel orientations of two m_s 's are equally probable is the average of the energy values 0 and $-K$ for the two cases considered separately. Hence $-\frac{1}{2}K(i,j)$ is the correct result. This argument, while perhaps not rigorous, is illuminating.

In Table III the energies of various valence states of a number of atoms and ions are estimated numerically by the use of the energy expressions in Table II taken in connection with available spectroscopic data. In many cases the results obtained can be regarded as only approximate for the well-known reason (among others) that the observed states do not fit Slater's formulas very well. For example, the observed ratio of the intervals $(^1S - ^1D)/(^1D - ^3P)$, which should be 3 : 2 for configuration s^2p^2 or s^2p^4 according to Slater's formulas, is actually much nearer 1 : 1. In spite of this difficulty, whose effects have, it is hoped, been minimized

by judicious averaging, etc. (cf. Table III), it seems probable that the final results given in Table III should represent fairly good approximations to the desired valence state energies.

In Table IV ionization potentials and electron affinities of a number of atoms are given, for both s and p electrons. Additional values can readily be obtained, when needed, by making use of the excitation energies of valence states, given in Table III. Most of the ionization processes given in Table IV correspond to removal of a valence electron so that if the atom is in state V_n , its positive ion is in state V_{n-1} . Similarly, many of the electron affinities correspond to addition of an electron in such a way as to go from an atom state V_n to a negative ion state V_{n-1} . This type of ionization potentials and electron affinities is what is needed for getting the electronegativity values of Table I. The electron affinities, it should be noted, are not accurate (probable errors ± 0.3 up to ± 1 volt) except in the case of the halogens: cf. Table IV, note†.